



Bangladesh University of Business and Technology (BUBT) Faculty of Engineering & Applied Sciences (FEAS) Department of Computer Science and Engineering (CSE)

LAB MANUAL

1	Program	B.Sc. Eng. in CSE													
2	Course Code	CSE 206													
3	Course Title	Digital Logic Design Lab													
4	Course Type	Core Course													
5	Credit Hour	1.50													
6	Course Objectives	The main objectives of this sessional course are to reinforce the theory presented in the CS Digital Logic Design course by allowing students to experiment with digital logic in a ha manner. And also understand both logical and electrical characteristics of digital design.													
7	Text Book	1. Digital Fundamentals, 9 th Edition, by Thomas L. Floyd													
8	Reference Book	1. Digital Design -With an introduction to the Verilog HDL, 5th Edition, by M. Morris Michael D. Ciletti													
9	Course Outcomes (COs)	Upon completing this course students will be able to: CO1: Demonstrate the working principles of different ICs & basic logic gates. CO2: Apply different IC's & logic gates for different digital logic circuits. CO3: Design and implement combinational circuits and sequential circuits.													
Mapping of COs to POs		Cours e Outco me	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO1 1	PO1 2	
		CO1	√												
		CO2									√				
		CO3			√										
CO No.		PO No.	Bloom's Domain / Level		Delivery Methods / Activities					Assessment Tools					
CO1		PO1	Cognitive / Understanding		Lectures, Lab Task, Assignment.					Lab Performance					
CO2		PO9	Affective/Respondin g		Lectures, Lab Task, Assignment.					Lab Performance and Final Exam					
CO3		PO3	Psychomotor / Guided Response		Lectures, Lab Task, Assignment.					Lab Performance and Final Exam					
10 Weekly Schedule															
Week		Lab	Task										Outcom e		
Week1		Lab1	To verify the Behavior of Logic Gates using Truth Table and familiarization with Digital Integrated Circuits										CO1		

Week2	Lab2	Implementation of Boolean Function using Logic Gates Lab Performance Evaluation	CO1
Week3	Lab3	Simplification of Boolean function by using Boolean Algebra and K- Map. Lab Performance Evaluation	CO2
Week4	Lab4	Construct and test various adder and sub-tractor circuits Lab Performance Evaluation	CO2
Week5	Lab5	Design, construct and test combination logic circuits like Majority Logic circuit, Parity Generator etc.	CO2
Week6	Lab6	Design of Code converters like Gray Code to Binary, Seven-Segment Display etc. Lab Performance Evaluation	CO2
Week7	Lab7	To check the operation of 2-to-4-line Decoder/3-to-8-line Decoder and design 4 to 16 line decoder.	CO2
Week8	Mid Term Week		
Week9	Lab8	To Design BCD (Binary Code Decimal) to 7-SEGMENT Decoder.	CO2
Week10	Lab9	To check the operation of active low decoder. Binary Coded Decimal encoder. Designing of octal to binary encoder and decimal to binary encoder. Lab Performance Evaluation	CO2
Week11	Lab10	Designing of 2-to-1, 4-to-1-line multiplexer and a quadruple 2-to-1-line MUX. Also implement 1-to-4 line de-multiplexer.	CO3
Week12	Lab11	Designing of 4-BIT Magnitude comparator and magnitude comparison using 74LS85 IC Implementing a Full Adder using (a) Decoder (b) Multiplexer	CO3
Week13	Lab12	Construct, test and investigate the operation of various flip-flop circuits, SR Latch, RS flip-flops, D flip-flop etc. Lab Performance Evaluation	CO3
Week 14	Lab13	Construction of Asynchronous Counters, Synchronous Counters and Registers. Lab Performance Evaluation	CO3
Week15	Lab Final Exam		